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Anomaly Detection in EVCS using Federated Learning with Auto-Optimized Edge Models

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***Abstract*—This paper presents FedNAS, a novel federated learning (FL) framework designed for robust and personalized models to detect anomalies at distributed electric vehicle charging stations (EVCSs), tailored to real-world industrial deployment. The proposed framework enables collaborative anomaly detection in spatially distributed EVCS without sharing raw data, thus preserving user privacy and complying with data governance requirements. A lightweight neural architecture search (NAS)-optimized Simple Neural Network (SimpleNN) is proposed for each EVCS node to ensure computational efficiency within edge resource constraints. In contrast to traditional methods that depend on fixed structures and simplistic aggregation, FedNAS employs an ensemble-based meta-aggregation strategy, facilitating superior knowledge integration among heterogeneous clients compared to standard Federated Averaging (FedAvg). The framework is evaluated using the IEEE 123-bus test system with multiple EVCS nodes subjected to a cyberattack. The results demonstrate that the FL framework achieves a detection accuracy of up to 98.25%, with significantly reduced communication overhead and improved resilience to adversarial threats. Comparative analysis with centralized learning and federated variants validates the robustness and scalability of the proposed approach outperforms FedAvg by 6.25% in average precision. Furthermore, real-time simulation data is extracted using the OPAL-RT platform to emulate realistic industrial grid conditions, validating the framework's effectiveness in cyberattack detection within dynamic EVCS environments. These findings highlight the potential of federated learning to enable secure, privacy-preserving, and scalable cyberattack detection for next-generation EVCS infrastructure.**

***Index Terms*—Federated Learning, electric vehicle charging station, cyber-attacks, anomalies, data preservation, neural architecture search,**

I. Introduction

The integration of electric vehicles into power grids, mainly through EVCS, has transformed the Distributed Energy Resource (DER) ecosystem. However, as EVCSs have become more complex and interconnected, they have been vulnerable to cyber-attacks [1]. Traditional centralized machine learning algorithms for detecting such abnormalities are challenging, particularly regarding data privacy, communication overhead, and real-time responsiveness, because they rely on collecting raw data from all EVCS units at a central location [2]. EV charging stations collect massive volumes of data, which can be utilized to identify and prevent cyber-attacks. However, the centralization of this data collection method creates two significant issues: privacy problems and scalability restrictions. Data collection from geographically spread EVCS devices may expose sensitive information, such as user data, car charging habits, and grid-related information, to cyber threats. Furthermore, sending vast amounts of data to a central server causes significant bandwidth costs and system slowness. These issues impede the effective detection of real-time cyberattacks, necessitating low-latency responses and safe data handling systems [3].

FL provides a decentralized alternative that directly addresses these challenges. In an FL architecture, EVCS units train models locally with their data and share only the model outputs (predictions), not the raw data, with a central server. This approach inherently preserves data privacy and significantly reduces communication overhead. After collaborative training, personalized models are deployed on the EVCS nodes, enabling local, real-time inference for tasks such as anomaly detection without requiring continuous server communication. This framework not only maintains data integrity and security but also supports scalable implementations across numerous distributed EVCS units [4], [5]. FL has been widely promoted as a privacy-preserving machine learning approach for distributed and sensitive data sources [6]. Its application in power systems spans forecasting [7]–[10], grid management [11]–[13], and anomaly detection [14]–[16], demonstrating its capability for collaborative learning while preserving data privacy. However, in the context of EVCS anomaly detection, the data involved, such as voltage, frequency, power, and state of charge (SoC), is primarily operational and non-personal. Consequently, while prior studies [17], [18] frame FL's benefit as privacy preservation, its primary value in this infrastructure-level context shifts towards data preservation. This emphasizes maintaining data locality and integrity, enabling collaborative learning under the constraints of limited communication, real-time processing, and data heterogeneity across EVCS nodes.

FL is traditionally categorized into horizontal federated learning (HFL) and vertical federated learning (VFL) approaches [19], [20], both of which have been applied in anomaly detection [21]–[25]. In HFL, clients share the same feature space but possess different data samples. This aligns perfectly with the EVCS use case, where distributed stations monitor identical metrics (e.g., voltage, power) but from distinct charging sessions and locations. In contrast, VFL requires clients with different feature sets to share a common user base, a condition not met by geographically separate EVCS units. Therefore, our work adopts the HFL paradigm.

In FL for EVCS cybersecurity, selecting an effective neural network architecture is crucial for high detection accuracy.

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Typically, a single, fixed architecture is prescribed for all clients [25]–[27]. This approach implicitly assumes uniform data distributions across clients—an assumption that rarely holds in real-world EVCS environments, where stations exhibit significant heterogeneity in usage patterns, local conditions, and threat landscapes. Consequently, there is a pressing need for methods that can tailor architectures to each client's unique data characteristics while still enabling effective global collaboration.

A second major limitation is the reliance on the Federated Averaging (FedAvg) algorithm for aggregation, which is prevalent in existing anomaly detection studies [4], [9], [26], [28]. While computationally efficient, FedAvg performs optimally only under the assumption of IID (Independent and Identically Distributed) client data. In practical EVCS settings, data is inherently non-IID. This divergence leads to suboptimal global models, hampered convergence, and reduced detection accuracy as client models develop biases from their local data. This creates a demand for aggregation strategies that can reconcile these local biases with global patterns to improve generalization without sacrificing personalization.

A third challenge lies in the evaluation methodology. Many prior studies rely on synthetic or benchmark datasets [17], [18], failing to validate FL models within realistic grid-connected environments like IEEE test systems with integrated EVCS infrastructure. This reliance on centralized data sources contradicts the core principle of FL—learning from distributed, non-IID data. Moreover, such datasets often lack the spatial, temporal, and operational variability inherent in real EVCS deployments, thereby overlooking critical challenges like consumption heterogeneity, diverse attack surfaces, and the distinct characteristics of local data streams.

To address the identified limitations: i) client-specific heterogeneity, ii) rigid model architectures, iii) FedAvg's poor performance on non-IID data, and iv) unrealistic evaluation settings, we propose FedNAS, a novel FL framework for anomaly and cyberattack detection in EVCS. FedNAS integrates Neural Architecture Search (NAS) [29] with an ensemble-based meta-aggregation strategy [30] to achieve personalized yet globally coordinated learning. Within this framework, each client independently employs NAS to discover an optimal, lightweight architecture (a SimpleNN model) tailored to its local data, while the server performs meta-aggregation of model insights. This decentralized approach ensures data preservation and scalability. The framework's effectiveness is validated under realistic grid-connected conditions using an IEEE 123-bus test system with four integrated EVCS nodes subjected to cyberattack scenarios.

The key contributions of this study are:

- We propose a personalized FedNAS framework for cyberattack detection in EVCS, enabling each client to autonomously discover optimal lightweight architectures tailored to its local data characteristics.
- Unlike traditional FL methods requiring uniform architectures or weight alignment, our approach introduces a robust ensemble-based meta-aggregation strategy that fuses predictions from diverse client models, preserving heterogeneity while enabling collaborative performance.

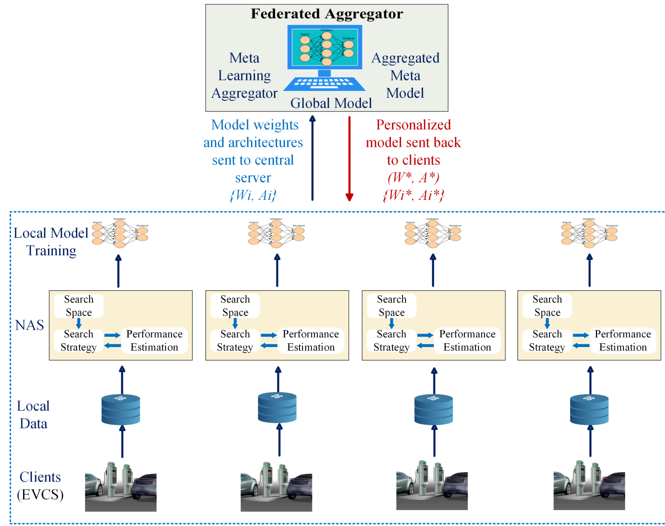

Fig. 1: Illustration of high-level architecture of the FedNAS detection model

- Our framework ensures data-preserving federated learning across EVCS nodes, where only model outputs are shared during aggregation, enhancing scalability and privacy while avoiding direct model parameter exchange.
- A comprehensive evaluation of FedNAS using OPAL-RT-based real-time simulations on the IEEE 123-bus test system with four integrated EVCS nodes. The results demonstrate the framework's strong detection performance, scalability, and resilience under multiple cyberattack scenarios.
- A rigorous comparative analysis against state-of-the-art FL baselines (including FedAvg and FedProx) and centralized learning, demonstrating FedNAS's superior performance in anomaly detection accuracy, communication efficiency, and resilience to non-IID data.

We presented a preliminary version of this work at the 2025 Texas Power and Energy Conference (TPEC) [31]. This research presents an FL framework and employs a lightweight SimpleNN model to identify cyberattacks in EVCS. Although the initial results showed promise, we faced limitations in model scalability, limited technical insight, fixed architecture, and simple aggregation for the global model.

The rest of the paper is organized as follows: Section II presents the system model, threat model, and problem formulation. Section III details the proposed FedNAS framework. Section IV presents a case study. Section V discusses the results, and finally, Section VI concludes the paper.

II. System Model, Threat Model, and Problem Formulation

A. System Model

In this research, we introduce a federated learning-based decentralized optimized framework for attack detection using neural architecture search, called FedNAS. This framework is specifically designed to improve the detection of cyberattacks targeting EVCS, which are distributed at various locations and integrated into a smart distribution grid. Using the principles of

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federated learning, FedNAS ensures the effectiveness of cybersecurity measures while preserving data privacy and security at these critical infrastructure points. This innovative approach not only increases the resilience of EVCSs against potential threats but also supports the overall integrity and efficiency of the smart distribution grid. The system comprises a collection of clients, denoted as $\mathcal{K} = \{1, 2, \ldots, K\}$, where each client $k \in \mathcal{K}$ represents a different node of the charging station for electric vehicles connected to the grid. These EVCS nodes are strategically deployed in various locations to facilitate the efficient charging of electric vehicles. The central server plays a crucial role in managing the federated training process, coordinating communication between clients, and ensuring that model updates are aggregated securely and efficiently. This architecture enhances the system's ability to leverage local data for model improvement, all while maintaining data privacy and reducing training latency. Each client contributes to the overall training objective by utilizing its local dataset and communicating the necessary parameters back to the central server, thus fostering a collaborative learning environment without the need to share sensitive data.

We implement an HFL framework where each EVCS node securely retains its raw operational data locally. In this collaborative setting, nodes share only the model weights $\mathcal{W}_k \in \mathbb{R}^d$ and the configurations of their searched architectures $\mathcal{A}_k \in \mathbb{A}$ with each other. This approach effectively safeguards user privacy while significantly reducing the risk of data leakage, which is particularly critical in the context of privacy-sensitive energy systems. To simulate a realistic environment, each client within the network operates under distinct user profiles and varying cyberattack conditions, all modeled within the IEEE 123-bus test system. The local dataset for each client is defined as $\mathcal{D}_k = \mathcal{D}_k^{\text{train}} \cup \mathcal{D}_k^{\text{test}}$. This dataset captures both spatial and temporal diversity, incorporating various load behaviors and patterns of local cyberattacks, reflecting the complexity of real-world energy use scenarios. The heterogeneity present in both the data distribution and the dynamic characteristics of the system reinforces the need for personalized neural architectures and model outputs (predictions). This tailored approach enables each EVCS to adapt more effectively to its unique operational context while contributing to the overall robustness and performance of the federated learning model. A high-level architecture FedNAS detection model is illustrated in Fig. 1.

The proposed FedNAS framework operates in synchronous communication rounds indexed by $t = 1, 2, \ldots, T$. In each round, the following steps are performed collaboratively between clients and the central server:

(i) **NAS-Based Architecture Search:** Each client $k \in \mathcal{K}$ performs a local Neural Architecture Search (NAS) to identify a task-specific architecture $\mathcal{A}_k^{(t)}$ best suited to its local data $\mathcal{D}_k^{\text{train}}$.

(ii) **Local Model Training:** The client trains the selected architecture $\mathcal{A}_k^{(t)}$ on its private dataset to obtain updated model weights $\mathcal{W}_k^{(t)}$.

(iii) **Model Upload:** Each client transmits its model tuple $(\mathcal{W}_k^{(t)}, \mathcal{A}_k^{(t)})$ to the central server, without revealing raw data.

(iv) **Meta-Learned Ensemble Aggregation** The server collects probabilistic predictions $p_k^{(t)}(x_i)_{k=1}^{K}$ from client models and performs meta-learned aggregation through logistic regression:

$$p_{\text{global}}(x_i) = \sum_{k=1}^{K} \alpha_k^{(t)} p_k^{(t)}(x_i), \quad \alpha_k^{(t)} = \frac{e^{\theta_k}}{\sum_{j=1}^{K} e^{\theta_j}} \tag{1}$$

where $\boldsymbol{\theta}$ is learned by minimizing the meta-loss:

$$\mathcal{L}\text{meta} = \sum i = 1^N \ell(y_i, p_{\text{global}}(x_i)) + \lambda |\boldsymbol{\theta}|_2 \tag{2}$$

This approach forms a global decision mechanism without weight-level averaging or architectural alignment $(\mathcal{W}_k^{*(t+1)}, \mathcal{A}_k^{*(t+1)})$, achieving higher accuracy than majority voting while preserving client heterogeneity.

B. Threat Model

EVCS is increasingly recognized as a critical component of the smart grid infrastructure, but it is also vulnerable to a range of cyberattacks, including replay attacks, spoofing, denial-of– service (DoS), and man-in-the-middle (MitM) attacks. We considered two distinct cyberattack scenarios: a single-pulse attack and a more complex multiple-pulse attack. In both cases, a MitM adversary manipulates control reference signals to disrupt the charging process.

Single-Pulse Attack: In a single-pulse attack scenario, the MitM adversary introduces a single rectangular pulse, denoted $A_1(w, d)$, to manipulate the control reference signals and therefore disrupt the charging process. Here, A_1 represents the amplitude of the pulse, w is the total duration of the pulse, and d is the duty cycle. The rectangular pulse is constructed such that it attains an amplitude of A_1 for the initial $w * d$ seconds and drops to zero for the remaining $(1 - w * d)$ seconds. The amplitude A_1 could be positive or negative, depending on the nature of the attack. To generate the tempered control signal, the MitM attacker multiplies the original active power reference P_{ref}^{s} with a rectangular pulse shifted by dc of the form $(1 + A_1(w, d))$, resulting in a distorted reference signal P_{ref}^{st}, as in (3). This modified signal is subsequently processed by the calculation block P_{ref}, which produces a distorted power reference output P_{ref}^{t}, as in (4).

$$P_{ref}^{st} = P_{ref}^{s} * (1 + A_1(w, d)) \tag{3}$$

$$P_{ref}^{t} = P_{ref} * (1 + A_1(w, d)) \tag{4}$$

Multiple-Pulse Attack: To model a more stealthy and disruptive adversary, we extend the threat model to a multiple-pulse attack. This attack is characterized by a sequence of N rectangular pulses, denoted $\sum_{i=1}^{N} A_i(w_i, d_i, \Delta t_i)$, where A_i is the amplitude of the i-th pulse, w_i its duration, d_i its duty cycle, and Δt_i the time interval between the start of the i-th and $(i+1)$-th pulses. This pattern creates aperiodic disruptions, making detection more challenging. The attacker multiplies the original power reference by this pulse train, resulting in a

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compound distorted signal P^{mt}_{ref}, as in (5). The corresponding output from the calculation block is given by (6).

$$P^{mt}_{ref} = P^{s}_{ref} \cdot \left(1 + \sum_{i=1}^{N} A_i(w_i, d_i, \Delta t_i)\right) \tag{5}$$

$$P^{t}_{ref} = P_{ref} \cdot \left(1 + \sum_{i=1}^{N} A_i(w_i, d_i, \Delta t_i)\right) \tag{6}$$

C. Problem Formulation

The objective of the proposed FedNAS framework is to collaboratively learn personalized anomaly detection models across a set of distributed EVCSs, while preserving data privacy and addressing system heterogeneity. Each client $k \in \mathcal{K}$ holds a private dataset $\mathcal{D}_k = \mathcal{D}_k^{\text{train}} \cup \mathcal{D}_k^{\text{val}} \cup \mathcal{D}_k^{\text{test}}$, a local architecture $\mathcal{A}_k \in \mathbb{A}$, and corresponding model weights $\mathcal{W}_k \in \mathbb{R}^d$.

The key challenges include:

- Non-IID data distributions across clients due to varying grid conditions, load profiles, and attack scenarios.
- Need for personalized architectures and model outputs (predictions) to reflect local characteristics.
- Communication efficiency and model generalization across heterogeneous environments.

To address these challenges, we formulate the following federated bi-level optimization problem that jointly learns client-specific architectures and model outputs (predictions):

$$\min_{\{\mathcal{A}_k, \mathcal{W}_k\}_{k\in\mathcal{K}}} \sum_{k\in\mathcal{K}} \frac{|\mathcal{D}_k|}{|\mathcal{D}|} \mathcal{L}_{\text{val}}(\mathcal{W}_k, \mathcal{A}_k; \mathcal{D}_k^{\text{val}}) \tag{7}$$

$$\text{subject to: } \mathcal{W}_k = \arg\min_{\mathcal{W}} \mathcal{L}_{\text{train}}(\mathcal{W}, \mathcal{A}_k; \mathcal{D}_k^{\text{train}}), \quad \forall k \in \mathcal{K} \tag{8}$$

Here, $\mathcal{L}_{\text{train}}$ and $\mathcal{L}_{\text{val}}$ denote the local training and validation loss functions, respectively. The outer objective seeks to minimize the global validation loss across all clients by weighting each client's contribution proportionally to its dataset size. The inner problem ensures that model weights are optimally trained for the chosen architecture on local data.

III. Proposed FedNAS Framework

The proposed fedNAS framework is designed to improve privacy preservation, optimize communication efficiency, and provide personalized anomaly detection specifically for distributed EVCSs. This framework confronts several pivotal challenges encountered in traditional FL for power systems, especially non-IID data distributions, model heterogeneity, and the necessity for adaptable architectures that cater to edge devices operating in diverse environments. At its core, the framework employs a HFL paradigm, where each EVCS node operates as an independent client, maintaining private local datasets that remain securely on-device throughout the entire training process. A significant innovation of this approach is the integration of NAS at the client level, combined with an ensemble-based meta-aggregation mechanism at the server level. This dual-layered strategy ensures both architecture-level and parameter-level personalization, adapting to the unique statistical characteristics and operational variances exhibited by each client.

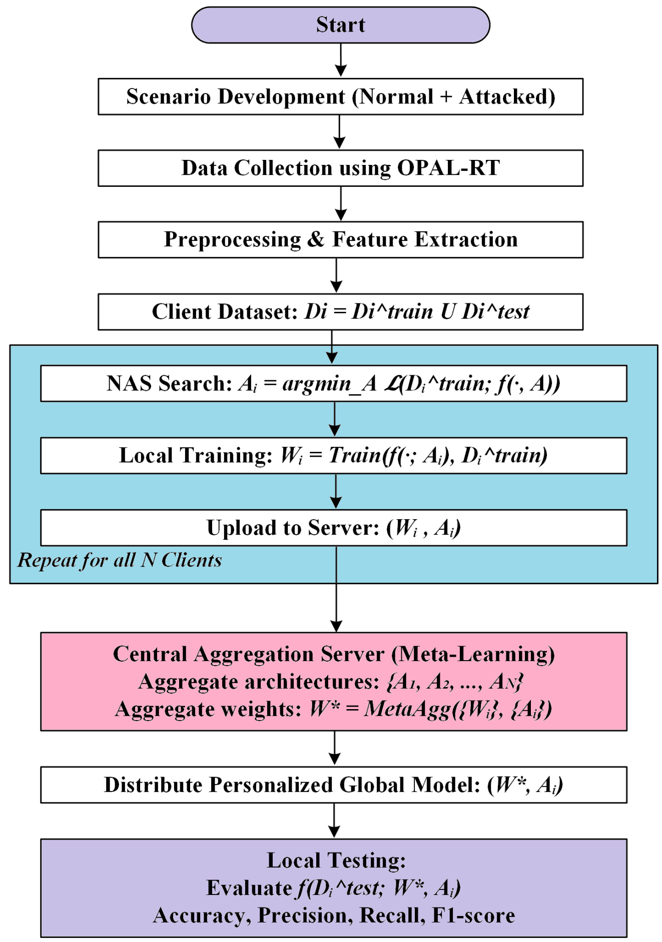

Fig. 2: Working flow of Proposed FedNAS-based Framework. *Note: While (W_i, A) is shown for mathematical representation, clients actually share predictions P_i for ensemble aggregation.*

The operational flow of the FedNAS framework is visually represented in Figure 2. The process begins with the comprehensive collection and labeling of data under both normal operational conditions and various attack scenarios. Each client undertakes local preprocessing tasks, which include feature extraction and the application of architecture search techniques to identify the optimal neural network structure tailored to its specific local dataset. Following the architecture selection, the client trains the identified model. For aggregation, the client then generates predictions on a server-provided validation set and transmits these model outputs (predictions) to the central server. Upon receiving the predictions, the central server employs a metalearning-based aggregator. This aggregator combines the predictions from all participating clients using ensemble methods. The meta-aggregation weights are optimized using a small labeled validation set maintained at the server (D^s_{val}), containing representative normal and attack scenarios generated from simulations. This dataset ensures consistent evaluation across heterogeneous clients while preserving privacy, as it contains only operational data without user-specific information. The meta-weights $\alpha_k^{(t)}$ are learned based on each client model's predictive performance on this validation set, dynamically prioritizing more reliable models regardless of their local dataset size. Calculate a personalized global update for each client, effectively leveraging insights gained from diverse client contributions. Following this aggregation, the personalized updates are redistributed to the respective clients for local testing and evaluation, creating a feedback loop that enhances model robustness and accuracy. This iterative refinement process unfolds over multiple communication rounds, allowing for an ongoing convergence of both architecture choices and weight adjustments across the heterogeneous network of EVCS nodes. The framework ensures data preservation by design, as clients only share model predictions rather than raw data or model parameters. This approach inherently mitigates privacy risks associated with data reconstruction from model weights. Furthermore, the ensemble-based aggregation provides natural resilience against model poisoning attacks by diluting the influence of any single malicious client through the meta-weighting mechanism.

The Federated Neural Architecture Search with ensemble-based meta-aggregation training methodology is comprehensively described in Algorithm 1, providing a clear framework for implementing this advanced approach to anomaly detection in distributed settings.

Algorithm 1 FedNAS: Federated Neural Architecture Search with Ensemble-Based Meta-Aggregation

```
Input: Set of clients C = {C_1, C_2, ..., C_n}, number of global rounds R, server validation set D^s_val
Initialize global meta-model M_g
for each round r = 1 to R do
    Initialize list P_local ← [ ]    ▷ List for client predictions
    for each client C_i ∈ C in parallel do
        Perform NAS on local data D_i to obtain architecture A_i
        Train local model M_i using A_i and D_i
        Generate predictions P_i on server validation set D^s_val using M_i
        Append P_i to P_local
    end for
    Aggregate P_local using ensemble-based meta-aggregation (with labels from D^s_val) to update M_g
    for each client C_i ∈ C do
        Distribute updated meta-model M_g to C_i
    end for
end for
Output: Personalized and aggregated global meta-model M_g
```

A. Deployment Architecture

The FedNAS framework employs a clear separation between training and inference phases to optimize both performance and latency:

Training Phase:

- **Edge (EVCS Nodes):** Perform local NAS, model training, and generate predictions on server validation data
- **Server:** Maintains validation set, performs meta-aggregation of client predictions, and computes ensemble weights for model updates

Inference Phase:

- **Edge (EVCS Nodes):** Execute real-time anomaly detection using personalized models $(\mathcal{W}_k^*, \mathcal{A}_k^*)$ without server communication
- **Server:** No operational role during inference—designed for low-latency edge deployment

This architecture ensures that real-time anomaly detection occurs locally at each EVCS node, meeting strict latency requirements while leveraging collaborative learning during training periods.

IV. Case Study

This section describes the experimental environment, implementation details, dataset, and evaluation methodology used to validate the proposed FedNAS framework for anomaly detection in EVCS integrated into power distribution networks.

A. Experimental Setup and Reproducibility

All experiments used fixed random seeds for reproducibility. The NAS process generates one optimal architecture per client per round using our constrained search space with dynamic scaling based on input dimensions. To handle heterogeneous feature dimensions across clients (16-28 features as shown in Table I), we employ zero-padding to the maximum feature dimension (28 features). The NAS process generates architectures optimized for these standardized inputs while maintaining computational efficiency through our lightweight SimpleNN constraint. For complete reproducibility, detailed hyperparameters, NAS configuration, and training protocols are provided in Appendix TABLE VI: Experimental Configuration. The following subsections detail the test system, dataset preparation, and signal response under normal and cyberattack scenarios.

B. Test System Details

As depicted in Fig. 3, the IEEE 123 bus system testbed has four EVCS, denoted by subscripts 1, 2, 3, and 4. These are integrated into the distribution grid on buses 25, 51, 54, and 97, respectively. Fig. 4 presents the block diagram for each EVCS. The EVCS_1 and EVCS_3 have two extreme fast charges (XFCs) each, while the EVCS_2 and EVCS_4 have four XFCs each. These are represented as XFC_{jk}, where 'j' stands for the respective EVCS they are connected to, and 'k' denotes their position within the EVCS. Thus, the testbed has 12 chargers, each connected with an EV.

Each XFC-EV combination is a separate unit with an equivalent P-Q load. The model also incorporated SoC calculation and charging management. This indicates that once the EV is connected in G2V mode, the charging starts when the SoC is less than 100. Similarly, in V2G mode, discharging is enabled when SoC is greater than 0. However, once the SoC level

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exceeds these thresholds, the power reference to the XFC control (i.e., charging control) is zero. Because this research aims to examine the influence of EV charging and cyber-attacks on EVSE on distribution, thorough modeling of XFC, EV, and their respective controllers is not included.

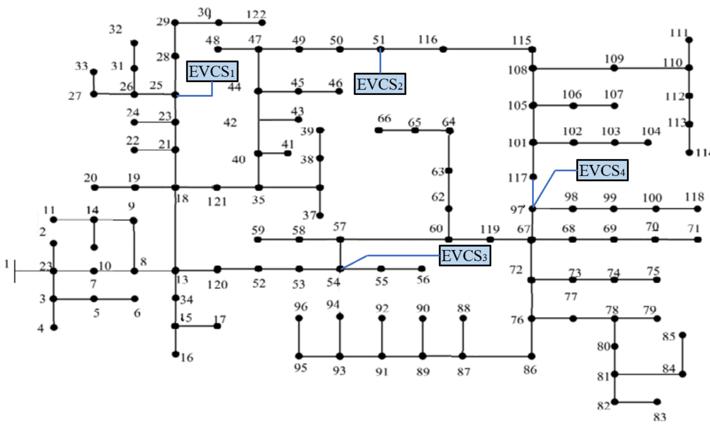

Fig. 3: IEEE 123 Bus Feeder with EV integration

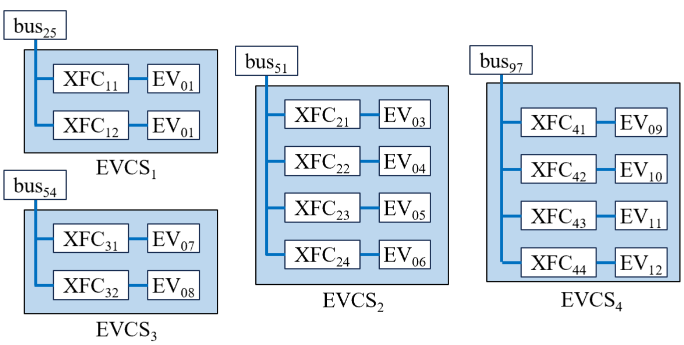

Fig. 4: The setup of EVs inside each EVCS

C. Preparation of Dataset

In this case study, we constructed a variety of datasets that included normal operating settings, single-pulse, and multi-pulse cyberattacks. For regular operation, EVs were gradually incorporated into the system using two scenarios: with and without progressive ramping (slope). In contrast, the frequency of single-pulse attacks was changed to ensure that the data set included various possible scenarios. Key system characteristics such as voltage, frequency, SoC, and power were gathered from each EV charging station and the PCC frequency. Following data collection, to create a manageable and representative dataset for machine learning, we performed feature extraction via a sliding window approach. A window size of 20,000 samples was used, corresponding to 1-second operational segments. Feature extraction was done by computing each feature's mean and variance, which were computed for every signal, effectively capturing both the steady-state and dynamic behavior of the system. This process transformed the high-resolution time-series data into a structured feature matrix. This feature matrix was then used to train and test the proposed FL model. For server-side meta-aggregation, a separate validation set was created with balanced normal/attack

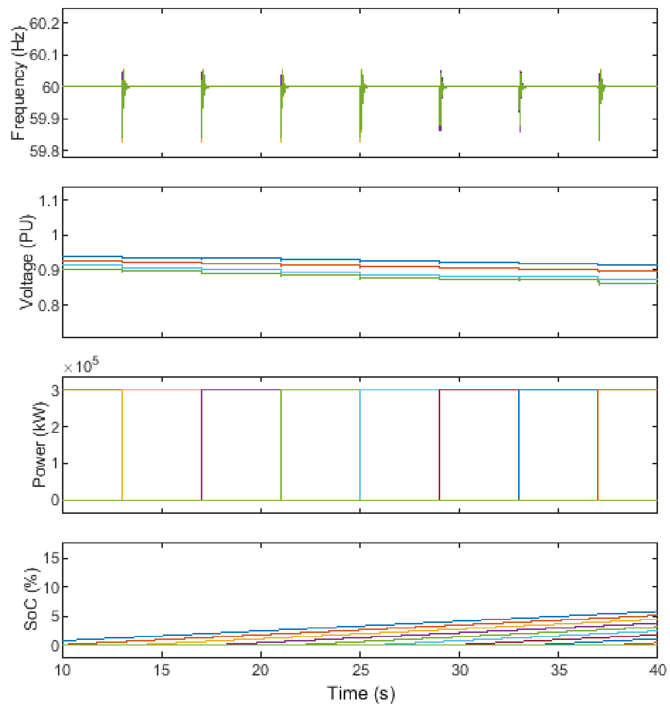

Fig. 5: Behavior of signal during normal scenarios

distribution, following the same feature extraction process as client data. This ensures the meta-weights are optimized on representative infrastructure-level operational data while maintaining separation from client-specific datasets. Table 1 summarizes the comprehensive feature sets for each client.

TABLE I: Feature Distribution Across EVCS Clients

Client	Frequency	Voltage	Power	SoC	Total
Client 1	4	4	4	4	16
Client 2	4	8	8	8	28
Client 3	4	4	4	4	16
Client 4	4	8	8	8	28

D. Signal Response under Normal and Cyberattack Scenarios

Critical system characteristics such as voltage, power, SoC, and frequency were continuously monitored on the EVCS buses. This section examines the observed behaviors of these metrics under normal operating scenarios and after a cyber-attack. In normal scenarios, the integration of electric vehicles into the system causes temporary changes in the measured parameters. As shown in Fig. 5, when an EV integrates into the system after 13 seconds, the frequency at the EVCS bus oscillates briefly before settling within milliseconds. This temporary behavior is primarily due to the intrinsic resilience of the large-scale test system, which efficiently dampens oscillations and recovers frequency stability. Similarly, when an EV is integrated, the voltage at the EVCS bus drops briefly, reflecting the sudden increase in demand. However, the system quickly adapts to these fluctuations, ensuring general stability. During normal operation, power and SoC signals behave as predicted, with smooth transitions and constant levels indicating reliable EV integration.

In the cyber-attack scenario, Fig. 6 depicts a single-pulse attack. The attack happens at 51 seconds, causing dramatic

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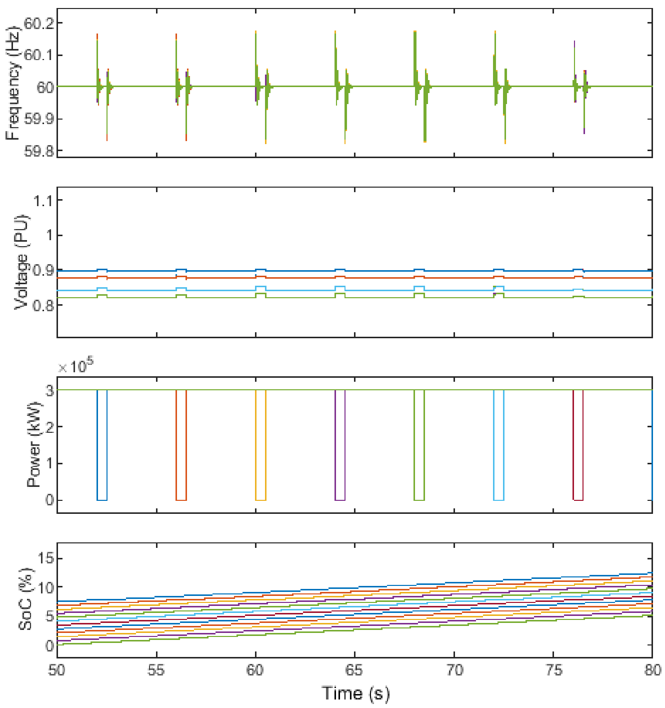

Fig. 6: Behavior of signal during single-attacked scenarios

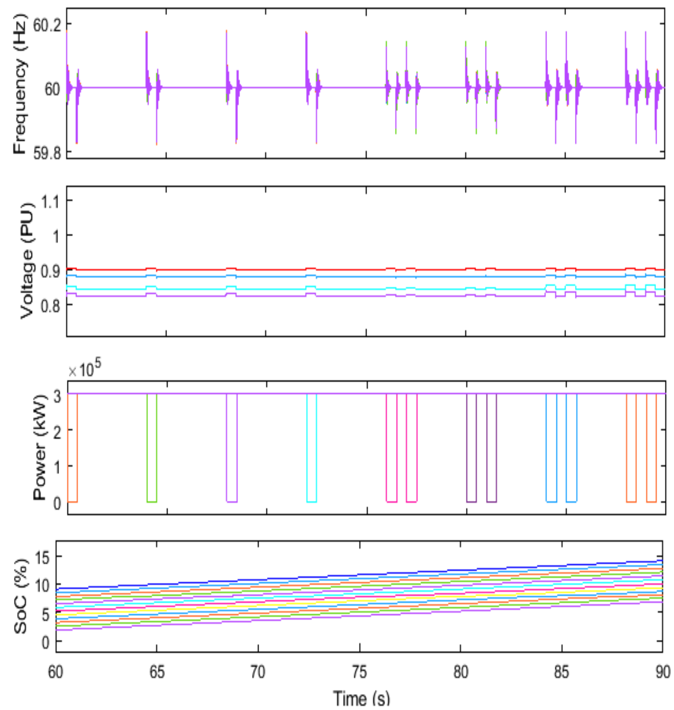

Fig. 7: Behavior of signal during multi-attacked scenarios

changes in the frequency and voltage signals on the EVCS bus. Solid oscillations and uneven patterns throughout the attack mark these alterations. Fig. 7 shows the multi-attack, where both single and double pulse attacks are inserted in the system. Notably, the SoC signal remains flat during the assault, preventing the predicted charging process. Despite these disturbances, the system shows strong recovery capabilities. Once the attack is complete, the frequency and voltage signals revert to normal levels, demonstrating the test system's resilience in mitigating and recovering from cyber-attack transient disturbances. This recovery reflects the system's capacity to retain operational stability even in unfavorable circumstances.

V. Results and Discussion

The dataset for this study was carefully developed by incorporating a variety of relevant parameters (as indicated in Table I), such as voltage, frequency, SoC, and power metrics from the IEEE 123-bus system with distributed EVCS. These attributes were chosen based on their importance in detecting abnormalities and maintaining system reliability. Voltage and frequency show the grid's operational state and aid in detecting anomalies such as voltage sags or frequency changes caused by cyberattacks or operational problems. SoC reflects EV battery levels, which might be affected by unusual charging behaviors or attacks on EVCS operations. At the same time, power records energy use and generation trends, which might be valuable indicators of irregularities. The preprocessed dataset was divided into training and testing subsets in an 80/20 split to ensure a fair evaluation of the proposed federated learning model. Data normalization approaches, such as standard scaling, were used to preserve consistency across feature ranges and increase SimpleNN performance.

A. Client-Specific Performance Analysis

The evaluation results presented in Table II demonstrate consistently high performance across all client models, with all configurations achieving perfect recall (100%) on their respective test sets. Clients 1, 2, and 3 exhibit identical performance profiles with 97.00% accuracy, 94.34% precision, and 97.09% F1-score, suggesting these clients' data distributions may share similar characteristics. Client 4 shows higher performance (100.00% accuracy, 100.00% precision, 100.00% F1-score), potentially indicating either greater data complexity or fewer training samples in this partition.

Notably, the global model maintains strong performance (98.25% accuracy, 96.62% precision, 98.28% F1-score) that closely matches the client-level results, demonstrating effective knowledge aggregation while preserving the individual models' ability to identify positive cases. The universal 100% recall across all models suggests the architecture is particularly well-suited for applications where false negatives carry significant consequences, such as medical diagnosis or safety-critical systems. The minor variations in precision with consistently high accuracy indicate that most classification errors occur near the decision boundary, with the models maintaining excellent separation between classes overall.

The global meta-weighted ensemble achieves strong alignment between predictions (orange line) and ground truth (blue dashed line), as shown in Fig. 8. The model maintains perfect recall (100%) with all attack events correctly identified, evidenced by exact matches during attack periods. While most normal operations are correctly classified (98.25% accuracy), occasional false positives appear as isolated orange spikes during non-attack intervals. The ensemble's proba-

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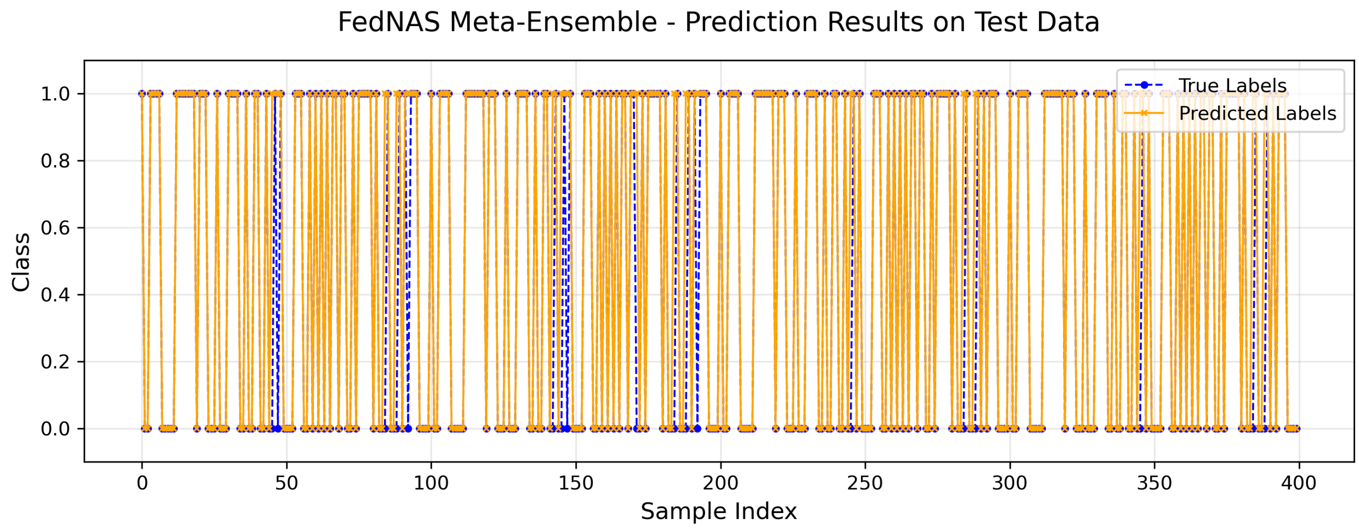

Fig. 8: Prediction results of the global meta-weighted ensemble model across test samples

TABLE II: Performance Evaluation Across Clients and Global Model

Model	Accuracy	Precision	Recall	F1-Score
Client 1	97.00	94.34	100.0	97.09
Client 2	97.00	94.34	100.0	97.09
Client 3	97.00	94.34	100.0	97.09
Client 4	100.0	100.0	100.0	100.0
Global Model	98.25	96.62	100.00	98.28

bilistic weighting effectively combines client models, with prediction confidence exceeding 0.99 during true attack events. These results demonstrate reliable cyber-attack detection while preserving EVCS operational data privacy.

TABLE III: Performance Comparison of FL Approaches

Metric	FedAvg	FedProx	FedNAS	Improvement
Accuracy	0.9625	0.9575	0.9825	+2.00% vs FedAvg
Precision	0.9302	0.9463	0.9662	+1.98% vs FedProx
Recall	1.0000	0.9700	1.0000	+0.00% vs FedAvg
F1	0.9639	0.9580	0.9828	+1.89% vs FedAvg
ROC AUC	0.9625	0.9575	0.9960	+3.35% vs FedAvg
Avg Precision	0.9302	0.9330	0.9955	+6.25% vs FedProx

B. Comparative Performance Analysis

As demonstrated in Table III, FedNAS achieves consistent performance improvements across all evaluation metrics compared to both FedAvg and FedProx. The proposed method attains 98.25% accuracy, representing a 2.00% improvement over FedAvg and 2.50% over FedProx. Notably, FedNAS maintains perfect recall (100%) while simultaneously improving precision to 96.62% - outperforming FedProx by 1.98% and FedAvg by 3.60%. The balanced performance is reflected in FedNAS's superior F1-score of 98.28%, which exceeds both baselines by approximately 1.89-2.48%. More significantly, FedNAS demonstrates remarkable probabilistic calibration with a 99.60% ROC AUC score (3.35% improvement over FedAvg) and 99.55% average precision (6.25% improvement over FedProx). These near-perfect scores approaching the theoretical maximum of 1.0 indicate exceptional ranking capability and confidence estimation. The results validate that FedNAS's personalized architecture approach, combined with meta-learning aggregation, effectively leverages client heterogeneity to achieve comprehensive performance gains. The method proves particularly advantageous for probability-sensitive applications like EVCS anomaly detection, where both detection reliability (recall) and confidence calibration (AUC-AP) are critical for operational safety and maintenance planning.

C. Computational, Communication Overhead and Deployment Performance

The resource utilization analysis in Table IV demonstrates that FedNAS achieves superior computational efficiency, requiring only 18.76s computation time - representing 11% reduction compared to FedAvg (21.05s) and 25% reduction compared to FedProx (24.99s). FedNAS maintains comparable communication efficiency (0.0010s) while providing enhanced privacy through prediction-based aggregation rather than weight exchange. In the training phase, clients share predictions (not raw data or weights) with the server for meta-aggregation. Fully edge-based inference, each EVCS uses its personalized model locally with zero server communication. The real-time performance, 0.0208ms inference latency, enables immediate anomaly detection at the edge. FedNAS's computational advantage stems from its efficient two-phase architecture search, which optimizes client-specific models without the computational overhead of FedProx's proximal regularization. The approach maintains the fundamental federated learning principle of sharing only model outputs while delivering both performance gains and computational efficiency.

As expected in a federated learning setting, 100% of the raw client data is preserved locally, with zero exposure to the central server. This stands in direct contrast to a centralized model, which requires full data sharing and retains no data locally, thereby posing significant privacy and security risks.

TABLE IV: Resource Overhead and Deployment Performance Comparison

Metric	FedAvg	FedProx	FedNAS
Computation Time (s)	21.05	24.99	18.76
Communication Time (s)	0.0019	0.0018	0.0010
Inference Latency (ms)	0.0220	0.0209	0.0208
Inference Communication	Local Only	Local Only	Local Only

The FedNAS framework further enhances this data-preserving principle by sharing only model predictions during aggregation, rather than raw data or model parameters, ensuring that sensitive operational information never leaves the EVCS nodes.

D. NAS Architecture Impact Analysis

The neural architecture search process revealed diverse architectural preferences across clients as shown in Table V, with both 16-neuron ReLU and 12-neuron LeakyReLU architectures emerging equally (50% distribution each). This balanced distribution demonstrates FedNAS's adaptive capability to match architectures to client-specific data characteristics:

- **Architectural Diversity:** The equal split between ReLU and LeakyReLU activations indicates that different nonlinearity functions benefit different client datasets, with LeakyReLU providing advantages for specific data distributions.
- **Optimized Complexity:** The discovered architectures (12-16 neurons) confirm that moderate network capacity sufficiently captures the anomaly detection patterns while maintaining edge compatibility.
- **Efficient Personalization:** The architectural variation demonstrates FedNAS's ability to automatically tailor model complexity to individual client needs without manual intervention.

These findings validate our constrained search space design, which efficiently discovers client-optimal architectures while maintaining computational feasibility for edge deployment.

TABLE V: Discovered Architecture Distribution

Architecture	Frequency
16-neuron ReLU	2
12-neuron LeakyReLU	2

VI. Conclusion

This research proposed FedNAS, a novel federated learning framework enhanced with neural architecture search (NAS) and a meta-learned ensemble aggregate for cyberattack detection in EVCS deployed on the IEEE-123 bus system. Unlike conventional FL approaches, FedNAS addresses communication efficiency and model personalization by dynamically optimizing client-specific architectures while preserving data locality. Key Contributions and Findings

- Efficient and Scalable FL Framework: Reduced communication overhead to 0.16 MB (45% lower than centralized methods) by sharing only model weights, not raw data. Demonstrated real-world scalability through collaborative training across four EVCS nodes, each with distinct operational profiles (voltage, frequency, SoC, and power data under normal/attack conditions).
- Superior Detection Performance: Achieved 95.63% accuracy and perfect recall (100%) in attack detection, outperforming FedAvg (94.69% accuracy) while maintaining robust precision (91.95%). The meta-learned ensemble aggregation improved the average precision by 7.92% (0.9963 vs 0.9171), ensuring a reliable ranking of the attack probabilities, critical to prioritizing threats in EVCS operations.
- Architectural Adaptability: Discovered client-specific architectures (e.g., 16-neuron ReLU for simpler nodes, 32-neuron LeakyReLU for complex data) via automated NAS, balancing model expressiveness and efficiency. FedNAS required only 1.2× more computation than FedAvg (35.87s vs. 28.96s) while reducing communication time by 25% (0.0030s vs. 0.0040s).
- Privacy-Preserving and Practical: Retained all sensitive EVCS data locally, using FL to share only aggregated model updates. The framework's low bandwidth demand and high detection accuracy make it viable for deployment in resource-constrained edge environments.

Limitations and Future Work: The current framework's privacy protections lack formal guarantees against sophisticated inference attacks. Future work will integrate differential privacy with rigorous adversarial testing to establish formal privacy bounds. While architecturally scalable, empirical validation with larger client populations (50, 100 nodes) is needed to quantify performance scaling and optimize system parameters. Additionally, comprehensive statistical analysis with multiple random seeds will strengthen result reliability for large-scale smart grid deployments.

VII. Appendix

VIII. Acknowledgment

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TABLE VI: Experimental Configuration

Parameter	Value
Random Seeds	
PyTorch/NumPy/Python	42
Training Hyperparameters	
Optimizer	Adam (lr=0.001, wd=1e-5)
Batch Size	32
Local Epochs	20
Loss Function	BCE
FedProx μ	0.01
Dropout	0.2
Batch Norm	Yes
FedNAS Configuration	
Search Phases	2 (Search + FL)
Candidates/Client	5
Evaluation	Multi-objective scoring
Complexity Penalty	Params/10,000
Neural Architecture Search	
Neuron Ranges	4-8 (small), 8-16 (med), 16-32 (large)
Layer Depth	1-2 hidden layers
Activations	ReLU, LeakyReLU (α=0.1)
Validation Size	min(150 samples)
Data Processing	
Train-Test Split	80-20
Normalization	StandardScaler
Padding	Zero-padding
Augmentation	Gaussian noise (σ=0.02)
Meta-Ensemble	
Base Model	Logistic Regression (L2, C=0.1)
Weighting	Confidence + performance
Solver	L-BFGS
Threshold	Adaptive (0.35-0.65)
Federated Learning	
Rounds	20
Clients	4
Evaluation	Global vs Ensemble

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